

2026/05/07

Quiz 4 Class 1 Solutions

Computer Security Fundamentals Spring 2026

Instructions: Answer the questions, **closed-book, closed-notes**. Time: 10 minutes

1. (**T**) Data Execution Prevention (DEP) prevents an attacker from executing code in memory regions that are marked as writable, such as the stack or the heap. (True/False, 2 pts)

2. (**A**) In a Return-Oriented Programming (ROP) attack, where is the sequence of "gadget" addresses typically stored by the attacker to direct the program's execution? (2 pts)

- A. On the program's stack.
- B. Inside the heap's metadata fields.
- C. In a read-only section of the program's binary.
- D. Within the CPU's instruction pointer (RIP) register.

3. (**C**) What is the primary purpose of a "Stack Canary" in a compiled program? (2 pts)

- A. To encrypt local variables so they cannot be read by an attacker.
- B. To provide root privileges to a process temporarily.
- C. To detect if a buffer overflow has overwritten the return address.
- D. To randomize the memory addresses of functions within an executable.

4. (Short Answer, 4 pts) A developer is designing a system with two components: a Network Listener (which receives untrusted data from the internet) and a Log Writer (which saves data to protected system directories). Following the Principle of Least Privilege, explain which component should be unprivileged and why.

Solution: The Network Listener should be unprivileged. Because it handles untrusted input, it is the most likely attack point for an attacker, and keeping it unprivileged ensures safety even if it is compromised. The Log Writer needs specific privileges to write to system folders but should be kept small and separate to minimize risk.